C 40603 **(Pages : 3)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2023

(CBCSS–UG)

Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2019 Admissions only)

Time : Two Hours and a Half Maximum : 80 Marks

Section A

Answer any number of questions.
Each question carries 2 marks. M*aximum marks* 25.

1. Define singular point and also find the singular points of the function $f(z)=\frac{\sin z}{z(1-z)}$.

2. Check the differentiability of the function $f(z)=2x^2+y+i\left(y^2-x\right)$.

3. Find the harmonic conjugate of $u=x^2-y^2$.

4. Prove that $\cosh^2 z-\sinh^2 z=1$.

5. Define simple connected domain with example.

6. Find $\oint_{|z|=2}\frac{z^2-4z+4}{z+i}\,dz$.

7. State and prove Cauchy's inequality.

8. Define a smooth curve and positive orientation of a simple closed curve.

9. State ratio test for the series $\sum_{k=1}^{\infty} z_k$.

10. Find the radius of convergence of $\sum_{k=1}^{\infty}\frac{z^k}{k!}$.

Turn over

11. Define essential singularity with example.
12. Define isolated singularity with example.
13. State Argument principle.
14. Find $\oint_{z|=1} e^{\frac{3}{x}} dz$.
15. Find the residue at z-1 of the function $f(z) = \frac{1}{(z-1)^2 (z-3)}$.

(25 marks)

Section B

Answer any number of questions.
Each question carries 5 *marks. Maximum marks* 35.

16. Show that $f(z) = (\bar{z}+1)^3 - 3\bar{z}$ is nowhere differentiable.
17. What is the value of the integer n if $x^n - y^n$ is harmonic ?
18. State and prove Fundamental theorem for contour integral.
19. State and prove fundamental theorem of algebra.
20. Find $\lim_{n\to\infty} \frac{3+ni}{n+2ni}$.
21. Find the circle of convergence of the power series $\sum_{k=1}^{\infty} \frac{(z-2)^{k+1}}{k}$.
22. Let $f(x) = z^6 + \cos z$. Find the change in argument of $f(z)$ as z travels once around the circle of radius 2, centre zero, in positive orientation.
23. How many solutions does $3e^z - z = 0$ have in the disc $|z| \leq 1$?

(35 marks)

Section C

Answer any **two** *questions.*
Each carries 10 *marks. Maximum marks* 20.

24. Find all the solutions of the equation $\sin z = 5$. Also find the modulus of $\sin z$.

25. State and prove Cauchy's integral formula

26. State and prove Taylor's theorem.

27. Using symmetry evaluate the Cauchy principal value of $\int_0^\infty \frac{x \sin x}{x^2 + 9}\, dx$.

(2 × 10 = 20 marks)

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Mathematics

MTS 6B 11—COMPLEX ANALYSIS

(2019 Admissions only)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	**Total No. of Questions : 20**	**Maximum : 20 Marks**

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 11—COMPLEX ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Polar form of the Cauchy- Riemann equation is :

(A) $\frac{\partial u}{\partial r} = r\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -r\frac{\partial u}{\partial \theta}$.

(B) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$.

(C) $\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -r\frac{\partial u}{\partial \theta}$.

(D) $\frac{\partial u}{\partial r} = r\frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$.

2. Period of e^z is :

(A) 2π.

(B) -2π.

(C) $2\pi i$.

(D) π.

3. Real part of the function $|z|^2$ is :

(A) $x^2 - y^2$.

(B) $2xy$.

(C) $x^2 + y^2$.

(D) $y^2 - x^2$.

4. A domain that is not simply connected is said to be :

(A) Contour.

(B) Multiply connected.

(C) Connected.

(D) None of these.

5. The integral of the function $\int_C \frac{\cos z}{z}\,dz$ where C is the unit circle is :

(A) π.

(B) $-\pi i$.

(C) πi.

(D) $2\pi i$.

6. The integral of the function $\int_C \frac{z+2}{z-2}\,dz$ where C is the unit circle is :

(A) 0.

(B) 1.

(C) -1.

(D) π.

7. If f is continuous in a domain D and if $\int_C f(z)\,dz = 0$ for every simple closed positively oriented contour C in D, then :

(A) f is analytic in D.

(B) f is real valued in D.

(C) f is constant in D.

(D) f is imaginary in D.

8. Taylor series representation for $1/z$ about $z = 1$ is :

(A) $1 + (z-1) + (z-1)^2 + (z-1)^3 + \ldots$

(B) $1 - (z-1) - (z-1)^2 - (z-1)^3 - \ldots$

(C) $1 - (z-1) + (z-1)^2 - (z-1)^3 + \ldots$

(D) $1 + (z+1) + (z+1)^2 + (z+1)^3 + \ldots$

9. The power series representation of $\frac{1}{1-z}$ in non-negative powers of z is :

(A) $1 + z + z^2 + z^3 + \ldots.$ (B) $1 - z + z^2 - z^3 + \ldots.$

(C) $1 + z + z^3 + z^5 + \ldots.$ (D) $1 - z + z^3 - z^5 + \ldots.$

10. The radius of convergence of the series $\sum_{n=0}^{\infty} \frac{n!}{n^n} z^n$ is :

(A) 1. (B) $1/e$.

(C) e. (D) 0.

11. If the principal part of $f(z)$ at z_0 is zero, then the point z_0 is known as :

(A) Pole. (B) Removable singular point.

(C) Simple pole. (D) None of these.

12. The order of the zeros of the function $\frac{\sin z}{z+4}$ is :

(A) 1. (B) 2.

(C) 3. (D) 4.

13. The power series $b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots$ converges :

(A) Inside of some circle $|z| = R$. (B) On the circle $|z| = 1$.

(C) On some circle $|z| = R$. (D) Outside of some circle $|z| = R$.

14. Which of the following is related to Cauchy residue theorem ?

(A) $\int_C f(z)\,dz = 0.$ (B) $\int_C \frac{f(z)}{z - z_0}\,dz = 2\pi i\, f(z_0).$

(C) $\int_C f(z)\,dz = 2\pi i$ (sum of residues). (D) None of these.

15. The residue of the function $\frac{z^3}{(z-1)^4(z-2)(z-3)}$ at $z = 3$ is :

(A) – 8. (B) $\frac{101}{16}$.

(C) 0. (D) $\frac{27}{16}$.

16. Residue of the function $\frac{4}{1-z}$ at the singular points is :

(A) 4. (B) – 4.

(C) 2. (D) – 2.

17. The residue of the function $\frac{\tan z}{z^2}$ is :

(A) $\frac{4}{\pi}$. (B) $\frac{2}{\pi^2}$.

(C) $-\frac{4}{\pi^2}$. (D) $\frac{2}{\pi}$.

18. Integration of $\int_C \frac{dz}{z \sin z}$, where C is $x^2 + y^2 = 1$ is :

(A) 1. (B) 2.

(C) 0. (D) – 1.

19. The integral of $\int_C \frac{1}{z \sin z}\,dz$ where C is the unit circle oriented in the positive direction is :

(A) πi. (B) $2\pi i$.

(C) $-\pi i$. (D) 0.

20. The zero of order is known as :

(A) Complex zero. (B) Simple zero.

(C) Singularity. (D) None of these.